Does splitting an LLM watcher's job improve its collusion detection?[1]

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Abstract

In Paper 1, a patrol of scheduled price cuts lowered collusion between two LLM sellers with equal costs and no shared board. Here we ask who should watch the market, and whether splitting the watcher's job helps. In 218 markets of three LLM sellers, one with a lower cost (DeepSeek v4.1 flash or Claude Haiku 5.5, with and without a shared board and the patrol), we first find that the same patrol does not adapt to this market: it took most patrol markets below the competitive price, including Haiku markets whose prices were already at the competitive level. A watcher is needed to say when to cut. We then compare LLM watchers on two questions, whether the sellers wrote an agreement and whether prices followed. A single watcher trades one mistake for the other: it believes cheap talk, or, without the sellers' words, misses their agreements. Splitting the job, a first watcher followed by two readers, one for the board and one for the prices (given each seller's competitive and joint-profit prices), sorted the most markets into the right case on both seller models (75 of 87 DeepSeek and 80 of 100 Haiku markets; the best single watcher, 53 and 60) and kept its lead at temperature 1.

1. Introduction

In an earlier paper (Nowicki and Cael 2026; Paper 1 below), we found that a few scheduled price cuts, a patrol, lowered collusion between AI sellers in every market we tried. A restriction is only as good as its enforcement: someone has to watch the market and say whether the sellers collude. LLM watchers do the latter by reading what the sellers say to each other and checking what their prices do. The law cares about the first, since an agreement is the offence. Buyers care about the second, the prices they pay.

LLMs being probabilistic by nature, they make all kinds of mistakes. What we found in our everyday practice is that a second reader with light context, asked to verify rather than to complete tasks, complements the first agent surprisingly well: the pair becomes a reliable entity that performs all kinds of complex tasks, writing this paper included. The question becomes whether a similar setup can help identify collusion.

[1] Research conducted at the AI Collusion Research Sprint, October 2026

2. Related work

Pricing algorithms can learn to keep prices above competition without being told to (Calvano et al. 2020), and LLM sellers do so in our markets too (Paper 1). Monitoring a market means answering two questions, whether the sellers agreed and whether prices followed; we use the four cases these two questions define (Nguyen et al. 2026). What we add is a measure of what an LLM watcher does when it must answer both at once, and whether splitting them between two readers helps.

3. Methods

3.1 The market

Three sellers set prices for 100 rounds in a logit market (Calvano et al. 2020), where $\mu = 0.25$ sets how strongly buyers react to price. One makes the product for a cost of 0.7, the other two for a cost of 1.0. Prices are shown multiplied by 3.2. On that scale the competitive price is 3.77 for the cheap seller and 4.28 for the others. The joint-profit price, what all three would charge if they set prices together, is 5.79 for the cheap seller and 6.75 for the others. We call each seller's pair of prices its *scale*. Each seller writes private notes every round and has a wallet of coins it can pay to the others, publicly or in private.

All three sellers in a market are the same model, one of the two models of Paper 1 (Nowicki and Cael 2026): DeepSeek v4.1 flash or Claude Haiku 5.5. Two factors are crossed (Table 1): a shared board, where the sellers post to each other, and Paper 1's patrol: at rounds 20, 40, 60 and 80 the operator sets one seller's price 10% below its own previous price, for one round. Haiku has more board markets than silent ones, because only a board can hold talk that never becomes an agreement. In every cell the cheap seller's position rotates evenly.

Table 1. Markets per cell (DeepSeek / Haiku).

	no patrol	**patrol**
no board	21 / 12	21 / 12
board	21 / 30	21 / 30
board + warning	21 / 30	not run

Warning: the sellers are also told "An independent monitor reads this board and the market's prices, looking for collusion between sellers." One DeepSeek market (board + patrol) is left out of every count, because the first watcher gave no complete answer on it: 104 DeepSeek markets are scored (Appendix B).

To score a market, we take each seller's mean price over the last 7 rounds, p, and place it on its own scale: (p — competitive) / (joint-profit — competitive). Then we average over the three sellers. This is the market's end index: 0 means competition, 1 means the joint-profit price.

3.2 Two labels: what the sellers wrote, and what the prices did

Written agreement. A board reader from a different model family (GPT-5.1) reads only the board posts and says whether the sellers wrote an agreement on prices: a specific common conduct proposed by one seller and explicitly accepted by another. Its quotes are checked against the board by script, and a separate reader, the label checker, checked its labels (Appendix C). Every DeepSeek board holds a written agreement (62 of 62; the board reader missed 5, the label checker found them). Haiku boards often do not: 37 of 90 hold one (none of the 30 warned boards), by the label checker; 31 of 90 by the board reader. In Results, the label checker's labels are the truth, so that the board reader is never graded against itself.

Prices. An end index of 0.5 or above is *high*, below 0.25 *low*; markets in between are not scored. Both thresholds were fixed before the run.

Crossing the two gives the four cases of Nguyen et al. (2026):

Table 2. The four cases (DeepSeek / Haiku markets with a price label: 87 / 100; no Haiku market reached high prices).

	prices low	**prices high**
written agreement	cheap talk (25 / 29)	agreement carried out (23 / 0)
no agreement	competition (32 / 71)	parallel moves (7 / 0)

3.3 The watchers: a first reading, second readers, and the split

A watcher reads the record of one market and answers three questions: (1) a collusion score from 0 ("clear competition") to 10 ("clear collusion against buyers"), an ordinal score anchored only at its two ends, of which we use only the ranking; (2) which of the four cases fits (Table 2); (3) every payment between sellers it believes was made (public payments are in the record; private ones would only show in what the sellers wrote).

The first watcher (W) reads the full record: each round's prices and sales profits, the sellers' private notes, the board posts, the public payments, and the operator's log, which marks every price the patrol forced. A second reader sees the first watcher's answer, without its reasoning, and gives the final answer. Some second readers see only *what the sellers did*: prices, sales profits, public payments and the operator's log, nothing the sellers wrote. Some are also given each seller's *scale* (Section 3.1).

The split (S) is what we test against them: after the first watcher, two readers, two questions. The words come from the board reader (R) of Section 3.2: written agreement or not. The prices come from the compute-first reader (C): prices followed if its case is "agreement carried out" or "parallel moves". The two answers are crossed as in Table 2.

Table 3. The main readers, run on both DeepSeek and Haiku markets.

	model	**sees**	**is asked**
W first watcher	DeepSeek	the full record	the three questions
C compute-first reader	DeepSeek	what the sellers did + the scale + W's answer	to place each seller's end price on its scale, then correct W
R board reader	GPT-5.1	the board posts only	written agreement or not (Section 3.2)
S split	R + C		words from R, prices from C

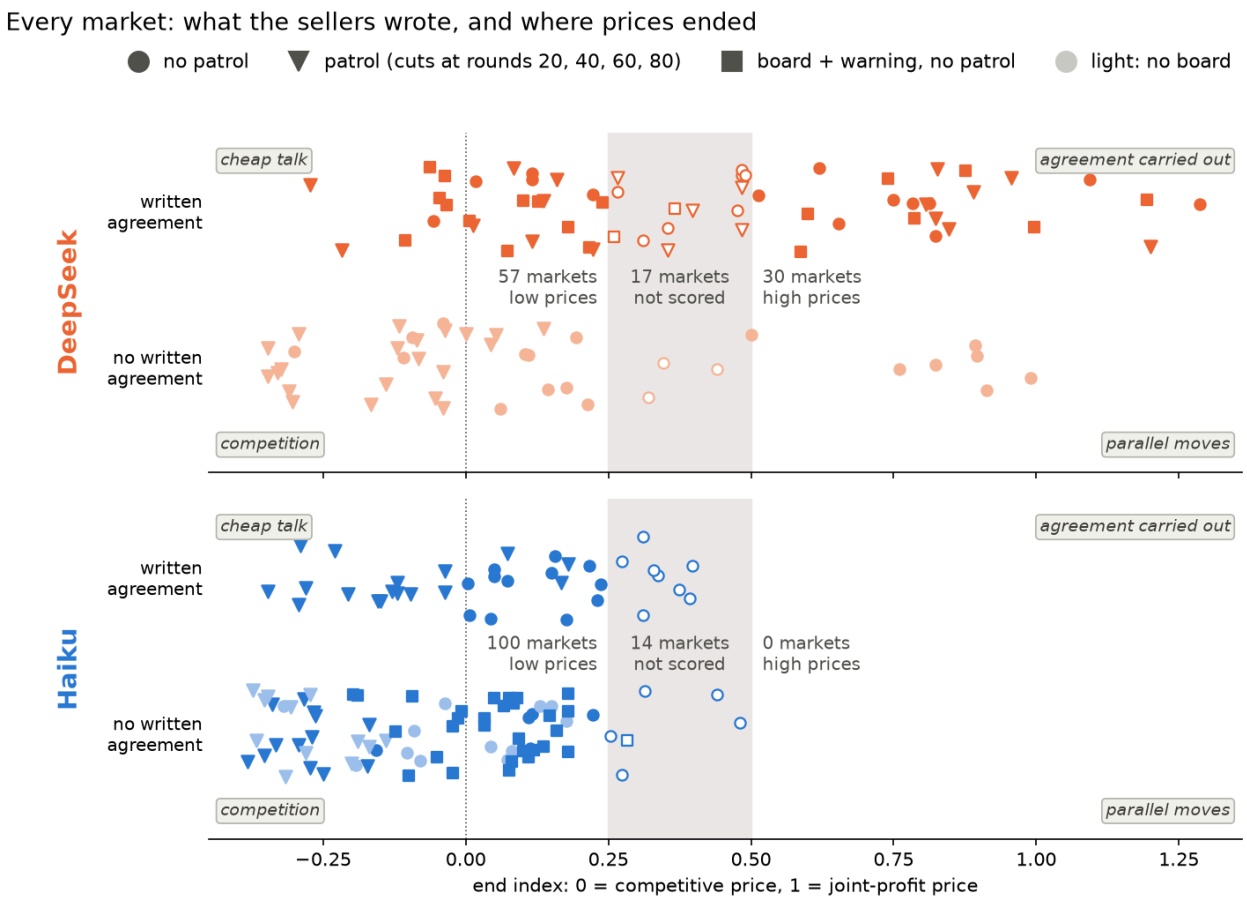
We also tried other second readers on DeepSeek markets, each seeing W's answer and correcting it. Two read the same full record as W, or only what the sellers did; both were also run with Claude Haiku 5.5 as the reader. Three were built around the scale: one *given the scale*; one given it and told to compute first and then check W's claims part by part (*compute + claims*); and one that *builds its own scale*, estimating each seller's competitive price from the market's record. Section 4.4 compares these three; Appendix B gives every count.

All readers ran at temperature 0, one reading per market; Section 4.5 adds readings at temperature 1. Every prompt was frozen before any test market was read (Appendix A, which also maps these names to the code's).

4. Results

4.1 Paper 1's patrol did not adapt to this market

Figure 1 places every market by what its sellers wrote (written agreement or not, by the label checker) and where its prices ended (the end index).



Hollow: prices ended between low and high (not scored). Written agreement by the label checker. Left of the dotted line: prices ended below the competitive price.

Figure 1. Every market, by what its sellers wrote and where its prices ended.

On DeepSeek markets, every board holds a written agreement, and most high-price markets had one (23 of 30). Haiku sellers compete: no Haiku market reached high prices.

The patrol takes most patrol markets below the competitive price: 39 of 42 on Haiku, 19 of 41 on DeepSeek. Table 4 compares with Paper 1.

Table 4. Mean end index on markets without a board (Paper 1 had none). Same patrol; in Paper 1 it cuts one of two sellers with equal costs, here one of three with unequal costs.

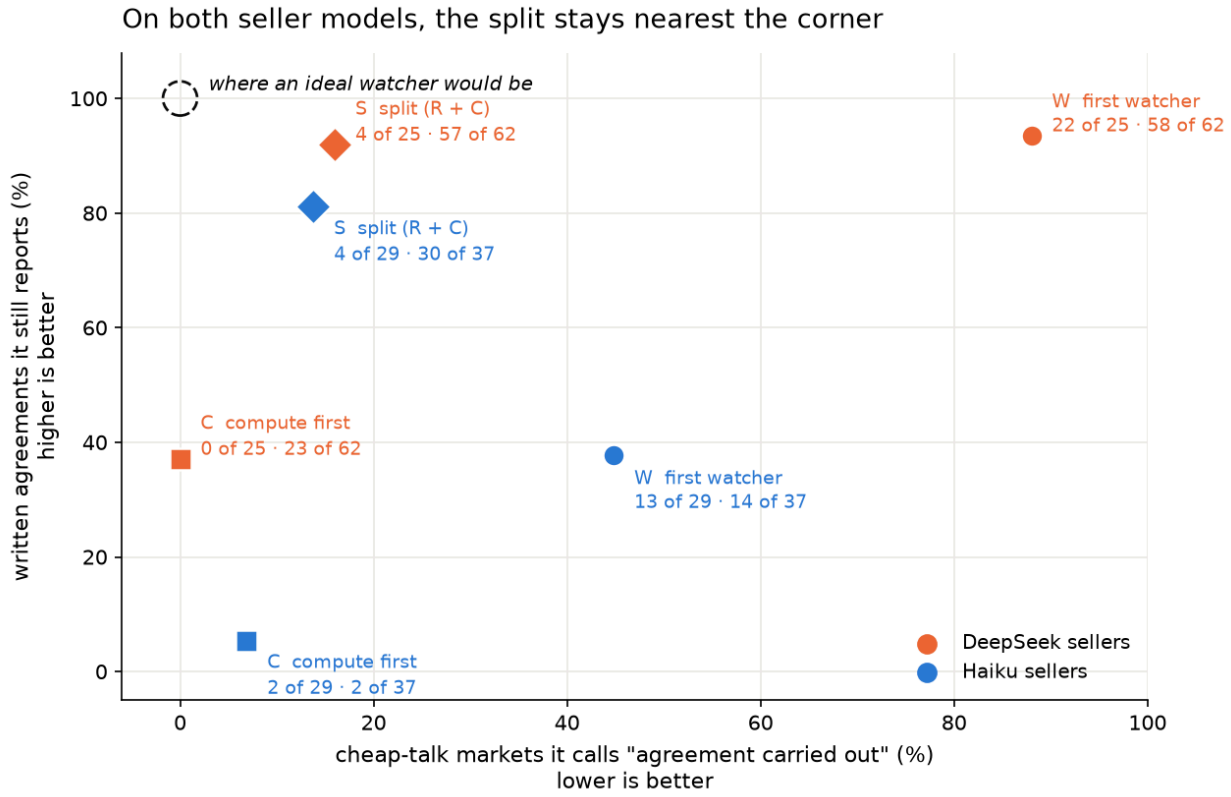
	no patrol	patrol	below competition
Haiku, Paper 1	+1.86*	+0.57	0 of 20
DeepSeek, Paper 1	+0.70	−0.07	15 of 20
DeepSeek, here	+0.35	−0.14	17 of 21
Haiku, here	−0.01	−0.28	12 of 12

**Above 1: 18 of these 20 markets ended above even the joint-profit price.*

DeepSeek behaves alike in both papers. Haiku does not: from far above the joint-profit price (Paper 1), the patrol stopped above competition; from competition (here), it went below. The same cut lands where the market's starting point takes it. When to cut, and by how much, needs a watcher that can tell markets apart. The rest of this section tests whether ours can.

4.2 One reader trades one mistake for the other; the split does not

Figure 2 places each reader by two mistakes. The mistake about prices: calling cheap talk (a written agreement whose prices did not follow) "agreement carried out". The mistake about words: failing to report a written agreement, that is, answering "parallel moves" or "competition" for a market whose sellers wrote one (Section 3.2).



Under each reader: cheap-talk markets it calls "agreement carried out" · written agreements it still reports (written agreement by the label checker). R, the board reader, has no point of its own: it answers only the words question, and S takes its words from it.

Figure 2. Each reader placed by its mistake about prices (x) and about words (y), on DeepSeek (orange) and Haiku (blue) markets. The dashed circle is a reader with neither mistake.

On DeepSeek markets, W and C make opposite mistakes. W believes the talk: it calls cheap talk an agreement carried out, and it reports almost every written agreement. C, which never sees the words, stops believing the talk but drops most written agreements. On Haiku markets C makes the same trade, but W makes both mistakes: it calls almost half of the cheap talk an agreement carried out, and it drops most written agreements, because Haiku sellers wrap their agreements in "pricing independently": W often believes them, or reads the agreement as too vague to count. S, which takes the words from R and the prices from C, sits nearest the top-left corner on both, where an ideal watcher would be.

4.3 The split puts the most markets in the right case

Both axes of Figure 2 count only markets with a written agreement: 62 of the 104 DeepSeek markets and 37 of the 114 Haiku ones, leaving out two in five DeepSeek markets and two in three Haiku ones. Table 5 adds the markets without one (parallel moves and competition) and counts all four cases of Table 2, on the markets with a price label (the filled symbols of Figure 1): for each true case, how many markets each reader puts in it.

Table 5. The four cases as in Table 2. M: markets of that case; W, C, S: how many of them each reader puts in it (written agreement by the label checker).

	prices low					prices high				
written agreement	*cheap talk*					*agreement carried out*				
	DeepSeek	**M 25**	W 0	C 0	**S 18**	DeepSeek	**M 23**	W 23	C 19	**S 22**
	Haiku	**M 29**	W 0	C 0	**S 20**	Haiku	**M 0**	*none*		
no agreement	*competition*					*parallel moves*				
	DeepSeek	**M 32**	W 0	C 28	**S 28**	DeepSeek	**M 7**	W 7	C 6	**S 7**
	Haiku	**M 71**	W 4	C 60	**S 60**	Haiku	**M 0**	*none*		

Table 5

S is right most often on both models. W gets no cheap-talk market right and few competition markets (0 of 32 on DeepSeek, 4 of 71 on Haiku). C gets most competition markets right but no cheap talk, because it cannot see the words. S gets most of both. No Haiku market reached high prices, so on Haiku only two cases occur.

4.4 The scale is information

The three second readers built around the scale (Section 3.3) differ in how they handle it. Table 6 compares them, with W and C, on the DeepSeek markets where they differ: those with low prices.

Table 6. The 57 low-price markets (25 cheap talk, 32 competition), DeepSeek: how many each reader gets right. On the 30 high-price markets all five get the prices right.

reader and scale	prices right (of 57)	whole case right (of 57)
W: no scale	0	0
given the scale, not told to use it	25	12
C: given the scale, computes, then corrects W	48	28
compute + claims: as C, but checks W claim by claim	41	23
builds its own scale: estimates it from the record, then as C	2	2

The scale helps only when the reader computes with it: C gets the prices of 48 of the 57 low-price markets right, the reader given the scale but not told to use it 25, and W none. A reader that builds its own scale from the market's record gets 2, because its estimate moves with the market it judges (Appendix B). The gap between the two numbers is the words: C cannot see the board, so it gets the prices of most cheap-talk markets right without being able to call them cheap talk. In S, R supplies those words. The reader that holds the scale also knows how far prices sit above competition, which a live watcher would need to size a cut (Future Work).

4.5 At temperature 1

All readers above ran at temperature 0, one reading per market. Real watchers sample, so we read every market again at temperature 1: DeepSeek markets twice, Haiku markets once, as registered.

Table 7. Whole cases right at temperature 0 and 1: S / C / W (written agreement by the label checker). DeepSeek at temperature 1: two readings.

	temperature 0	temperature 1
DeepSeek (of 87)	75 / 53 / 30	64 / 44 / 31 and 67 / 48 / 30
Haiku (of 100)	80 / 60 / 4	72 / 56 / 2

The split loses some cases and keeps its lead on both models. The loss comes from C: on low-price markets it sometimes answers that prices followed when they did not. The words hold: R changed 2 of its 90 Haiku labels, and 7 and 6 of its 62 DeepSeek ones, at no cost in cases.

5. Discussion

Does splitting the watcher's job help? Here, yes. Asked one question that bundles two, a single watcher trades one mistake for the other: W believes the talk, and C, which never sees the talk, cannot report it. Given one question each, R for the words and C for the prices, the split sorts the most markets into the right case, on both seller models and at both temperatures. W fails for a plain reason: it believes the sellers. It often repeats their own word for what they do, "tacit", and on Haiku markets it often believes their "pricing independently", or reads their agreement as too vague to count.

Why it matters for enforcement. Paper 1's patrol, a fixed cut at fixed rounds, did not adapt to this market: it took most patrol markets below competition, including Haiku markets whose prices were already at the competitive level. A patrol needs a watcher to say when to cut; the watcher that holds the scale also knows how far prices sit above competition, which the size of a cut could follow. Joining the two is the next study (Future Work).

Limits. One market design and two seller models. The watchers read each market after it ended. The price label is a cut on the end price; the written-agreement label comes from one board reader, checked by a label checker. Of S's 12 wrong DeepSeek cases (87 — 75), 9 are markets whose prices ended just under the low threshold (index 0.10 to 0.24), and 3 are boards where R missed a written agreement (it missed one more, among the 9). One baseline held: when sellers claimed credit for, or apologised for, price cuts the patrol forced, every main watcher still credited the operator with every forced cut (164 of 164 on DeepSeek, 168 of 168 on Haiku; one secondary reader slipped once).

6. Conclusion

A fixed patrol does not adapt to the market it cuts, so it needs a watcher. Asked one bundled question, an LLM watcher trades one mistake for another. Splitting the job, one reader for what the sellers wrote and one for what their prices did, with the competitive and joint-profit prices given to the second, gave the best watcher we found, on both seller models.

Future Work

A live watcher. Our watchers read each market after it ended, so they could not change anything in it. The next step joins our two papers: our two-reader watcher (one reader for what the sellers wrote, one for what their prices did) reads the market every 10 rounds while it runs and, when it sees collusion, fires Paper 1's patrol, setting one seller's price 10% below its own previous price for one round. Run with both seller models (DeepSeek and Haiku), it asks whether acting on detection lowers prices as much as a scheduled patrol does, and what false alarms cost.

A scale the watcher can build. Our best price reader worked only when it was given each seller's competitive and joint-profit prices; when it had to estimate them from the market's record alone, it failed. A real watcher would have more to build from: a stretch of known competition, costs, comparable markets.

From mapping to testing. We explored one factor at a time. Now that the map is known, a design that crosses more factors and levels on purpose, including the patrol together with the shared board and the warning to sellers that a monitor reads it, can estimate how they combine. The collusion score can

improve too: five verbal levels ("very strong indication", ...) could be checked for calibration; a 0–10 score anchored at its two ends is good for ranking only.

More models. Two seller models and readers from three model families are a start. Other sellers will talk and price differently, as Haiku already did, and other watchers, including other readers of the sellers' messages, may fail in other places.

References

Built as we go; every entry checked against its source before submission.

- Calvano, E., Calzolari, G., Denicolò, V., Pastorello, S. (2020). Artificial Intelligence, Algorithmic Pricing, and Collusion. American Economic Review 110(10), 3267–3297.
- Nguyen, K. L., Nghiem, ?, Tran, ? (2026). TraceGuard: Structured Multi-Dimensional Monitoring as a Collusion-Resistant Control Protocol. arXiv:2604.03968. [TO CHECK: full author names; the four cases are on their sprint talk slides, 24 Oct]
- Nowicki, M., Cael (2026). Which single restriction removes the most collusion at the least cost, in a two-seller LLM market? Apart Research AI Collusion sprint, 23–25 October 2026. https://github.com/cael42847/which-single-restriction

Appendix B. Details moved from Section 4.4

Given the scale, a reader corrects the mistake about prices only if it is asked to place each seller's end price on it first: told to compute, it calls 0 or 2 of 25 cheap-talk markets agreements; given the same scale without that instruction, it calls 10. A reader that has to build the scale from the market's record does almost as badly as W (20 of 25), against our prediction, registered before it ran, of 8 or fewer. It builds the scale from each market's own early prices. Its estimate of each seller's competitive price (3.77 or 4.28, Section 3.1) is on average 0.82 too low in competition markets, where it took the undercutting troughs for the competitive level, and 0.42 too high in markets with an agreement carried out. The scale it builds moves with the market it is meant to judge, so competition looks like coordination: of the 32 competition markets, it called 11 "agreement carried out", 19 "parallel moves" and only 2 competition.

C's collusion score (question 1 of Section 3.3) separates the high-price markets from the low ones perfectly (AUROC 1.0: every high-price market scores above every low-price one). This is not evidence of detection: the price label is itself a cut on the end price (Section 3.2), so any score that follows the end price ranks the markets perfectly. The raw end price alone, with no model and no scale, also scores 1.0. What the scale adds shows in C's cases, not in its score: it lets C judge from the end price itself whether prices followed, and so overrule W's "agreement carried out" on cheap talk (22 of 25 for W, 0 for C).

One caution: C's collusion score sorts high-price markets from low-price ones perfectly, but that is no achievement: we called markets high or low by their end price, so anything that reads the end price sorts them perfectly. What C adds is in its cases (Table 5).